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Software Engineering-I

Provided by **VU Answer**

Q1) Subtypes of Loops

Answer:

1. For Loop
2. While Loop
3. Do-While Loop

Q2) Alternatives of "is prefix"

Answer:

- has
- can
- does

Q3) Intel or Motorola Architecture

Answer:

Intel and Motorola are two different types of microprocessor architectures.

Intel uses the little-endian format, where the least significant byte is stored first.

Motorola uses the big-endian format, where the most significant byte is stored first.

Q4) Cyclomatic Complexity

Answer:

Cyclomatic complexity measures the number of linearly independent paths through a program's source code. It helps to determine the complexity of a program.

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Q5) Components of Client-Server Architecture

Answer:

- Client
- Server
- Network

Q6) Get/set or Is prefix uses

Answer:

- **Get:** Used to retrieve the value of a property.
- **Set:** Used to assign a value to a property.
- **Is:** Used for boolean properties to check a condition (e.g., isAvailable).

Q7) Boolean equivalence of English letter "and"?

Answer:

The boolean equivalent of "and" is &&.

Q8) Message types of sequence Diagram

Answer:

1. Synchronous Message
2. Asynchronous Message
3. Return Message
4. Create Message
5. Delete Message

Q9) UML and their behavior ki types

Answer:

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UML (Unified Modeling Language) is a standardized modeling language used in software engineering to visualize the design of a system.

The main types of behavioral UML diagrams include

1. Use Case Diagram
2. Sequence Diagram
3. Activity Diagram

Q10) Identify below given statement as correct or incorrect, justify your answer in either case. "User requirements are written from a system perspective".

Answer:

Incorrect because user requirements should be written from the user's perspective to ensure they meet the user's needs and expectations.

Q11) In Data-Centered (Repository) model, sub-systems need to exchange information and data. Write the two ways in which this exchange of the data can be handled?

Answer:

Data exchange can be handled by

Direct method calls: Sub-systems call each other directly to exchange data.

Shared data repository: Sub-systems read from and write to a common data store.

Q12) What is the appropriate use of the term "initialize"?

Answer:

"Initialize" means to set a variable to a starting value before it is used in a program.

Q13) Statement: `int myBooks[5] = {1, 2, 3, 4, 5}` Identify the coding mistake and correct it.

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Answer:

Correct Statement

int myBooks[5] = {1, 2, 3, 4, 5}; (Add a semicolon at the end).

Q14) Two main documents produced during requirement engineering phase, are Requirement Statement and Requirement Specification. You are required to write down any five characteristics of good requirements statement document.

Answer:

1. Clear
2. Concise
3. Consistent
4. Complete
5. Verifiable

Q15) A software house wants to launch an E-commerce website, but they want to perform software testing. They choose unit testing for the website. Is unit testing helpful in a small-scale organization elaborate with a reason? Enlist any two benefits.

Answer:

Yes, unit testing is helpful. It ensures that each part of the website works correctly. Benefits include

Early Bug Detection: Issues are found early in the development process.

Easier Maintenance: Easier to make changes and updates.

Q16) A Student class has following attributes and operations. Convert the "Student" class in the form of diagram using UML notation for classes. Attributes a. Roll no. b. Name c. Subject Operations a. AttemptQuiz() b. SubmitAssignment() c. PayFee()

Answer:

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```
-----  
|      Student      |  
|-----|  
| - rollNo : int    |  
| - name : string   |  
| - subject : string|  
|-----|  
| + attemptQuiz()   |  
| + submitAssignment()|  
| + payFee()        |  
-----
```

Q17) "is" prefix three alternatives

Answer:

- has
- can
- does

Q18) Three- tier architecture Layers name

Answer:

1. Presentation Layer
2. Business Logic Layer
3. Data Access Layer

Q19) Name the UML diagrams.

Answer:

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1. Class Diagram
2. Use Case Diagram
3. Sequence Diagram
4. Activity Diagram

Q20) while a

```
{  
while b  
c  
d  
}
```

What would be the test technique for this code?

Answer:

Use path testing to ensure all paths through the code are executed.

Q21) Correct the statement

"In the telecommunications industry, loss of service is known as lack of service"

Answer:

Correct Statement

"In the telecommunications industry, loss of service is known as service outage."

Q22) How does difference in bite ordering between Intel and Motorola architecture affect data exchange?

Answer:

Intel uses little-endian, where the least significant byte is stored first. Motorola uses big-endian, where the most significant byte is stored first. This difference affects how data is interpreted and exchanged between systems.

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Q23) What is the maximum character limit for a single line in c++?

Answer:

The C++ standard does not define a maximum character limit for a single line, but practical limits are imposed by the compiler.

Q24) Name the two types of components in collaboration diagram.

Answer:

1. Objects
2. Messages

Q25) Which line is used to organise the following units vertical or horizontal in a sequence diagram. Objects, events, message flow, activation box, object lifeline

Answer:

Use a vertical line to organize objects, events, message flow, activation box, and object lifeline.

Q26) for (i=0; i<100; i++) { //1

if (i<a)

c=c+1; //2

else

c=c-1; //3

b=b+c; //4

} //5

Draw a flow graph and find cyclomatic complexity.

Answer:

Cyclomatic complexity for the given code is calculated using the formula: $M = E - N + 2$

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- E is the number of edges.
- N is the number of nodes.

Flow graph and complexity

Nodes: 5

Edges: 6

Cyclomatic complexity = $6 - 5 + 2 = 3$

```
Q27) for(i=0; i<N; i++){ //1
if (condition 1)
//Do something here //2
else
//Do something here //3
//Something here //4
}
```

What is the number of possible paths? Also identify those paths. (N=2)

Answer:

For N=2

- Path 1: {1, 2, 4}
- Path 2: {1, 3, 4} Total paths: 2

Q28) Break this statement into simple set of statements, to make it easier to understand:

```
*x += (*xp=(2*k < (n-m) ? c[k+1] : d[k--]));
```

Answer:

Simplified

1. if ($2*k < (n-m)$)

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2. `*xp = c[k+1];`
3. `else`
4. `*xp = d[k--];`
5. `*x += *xp;`

Q29) UML PROVIDE TWO DIFFERENT MECHANISM? NAMES?

Answer:

Structural Diagrams

Show the static structure of the system (e.g., Class Diagram, Object Diagram).

Behavioral Diagrams

Show the dynamic behavior of the system (e.g., Sequence Diagram, Activity Diagram).

Q30) CLIENT-SERVER MODEL 4 CONFIGURATIONS NAMES.

Answer:

1. Two-Tier Architecture
2. Three-Tier Architecture
3. N-Tier Architecture
4. Peer-to-Peer (P2P)

Q31) ARITHMETIC AND LOGICAL RIGHT SHIFT OPERATOR

Answer:

Arithmetic Right Shift: Preserves the sign bit (e.g., `>>` in some languages).

Logical Right Shift: Shifts bits and fills with zeros (e.g., `>>>` in some languages).

Q32) Boolean equivalence of English letter "and"?

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Answer:

The boolean equivalent of "and" is &&.

Q33) Message types of sequence Diagram

Answer:

1. Synchronous Message
2. Asynchronous Message
3. Return Message
4. Create Message
5. Delete Message

Q34) Write different types of coverage used in White box testing?

Answer:

1. Statement Coverage
2. Branch Coverage
3. Path Coverage
4. Condition Coverage

Q35) Special characters that are avoided in multi-platform environment?

Answer:

- Backslash (\)
- Colon (:))
- Asterisk (*)
- Question Mark (?)
- Double Quote (")

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Q36) Write try catch syntax?

Answer:

```
try {  
    // Code that might throw an exception  
} catch (ExceptionType e) {  
    // Code to handle the exception  
}
```

Q37) Software architecture is a set of elements which have a certain form write name of these elements?

Answer:

- Components
- Connectors
- Configurations

Q38) Write name of any five GOF Design Pattern Format?

Answer:

1. Singleton
2. Observer
3. Factory Method
4. Adapter
5. Decorator

Q39) Write Checklist for static analysis?

Answer:

1. Code Conventions
2. Syntax Errors

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3. Code Complexity
4. Dead Code
5. Security Vulnerabilities

Q40) According to a famous software engineer martin fowler, “any fool can write code that computers can understand, good programmers write code that humans can understand.”

What do you understand by above statement?

Answer:

Martin Fowler's quote means that while anyone can write code that a computer can run, skilled programmers write code that is easy for other humans to read and understand. This makes the code maintainable and easier to work with in the future.

Q41) What do you understand by the term effective testing? which software document is

considered as the base document to start the testing phase of software?

Answer:

Effective testing ensures the software works correctly and meets requirements. The base document for testing is the Requirements Specification.

Q42) Write name of krutchen's architectural view models.

Answer:

1. Logical View
2. Development View
3. Process View
4. Physical View
5. Scenarios (Use Cases)

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Q43) During the testing phase of software, when are two Tests considered to be equivalent?

Answer:

Two tests are considered equivalent if they test the same aspect of the software and would produce the same results, even if they are different inputs or scenarios.

Q44) Coverage types used in white box testing

Answer:

- Statement Coverage
- Branch Coverage
- Path Coverage
- Condition Coverage

Q45) What is documented in requirements document view.

Answer:

The requirements document details what the software should do, including all functional and non-functional requirements.

Q46) Kirchhoff's architectural view model

Answer:

This is the 4+1 view model which includes Logical, Development, Process, Physical views, and Scenarios.

Q47) Draw control flow diagram of 2 scenarios

Answer:

A Control Flow Diagram visually represents the flow of control in a program, including decisions and paths. (Diagrams not shown due to text format limitations).

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Q48) Two Principal of Unit Testing

Answer:

1. Isolation
2. Repeatability

Q49) Define Defects

Answer:

Defects are flaws or bugs in the software that cause it to produce incorrect or unexpected results.

Q50) Name White Box testing coverage scheme

Answer:

1. Statement Coverage
2. Branch Coverage
3. Path Coverage
4. Condition Coverage

Q51) Cyclomatic Complexity

Answer:

Cyclomatic Complexity measures the number of linearly independent paths through a program's source code. It's calculated as $M = E - N + 2P$ where E is the number of edges, N is the number of nodes, and PPP is the number of connected components.

Q52) Advantage and disadvantage of client-server model

Answer:

Advantages

- Centralized resources and data.

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- Easier maintenance and management.

Disadvantages

- Single point of failure.
- Network dependency.

Q53) Define White box testing, Black box testing and Pattern phase

Answer:

White Box Testing: Tests internal structures or workings of an application.

Black Box Testing: Tests the functionality without looking at the internal code.

Pattern Phase: Refers to the design pattern phase in software development.

Q54) Unit testing two characteristics

Answer:

1. Each test is independent.
2. Tests should be run automatically.

Q55) OO model ki layer k name.

Answer:

1. Class Layer
2. Object Layer
3. Method Layer
4. Message Layer

Q56) Client server model

Answer:

A model where clients request services, and servers provide them.

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Q57) Write the name of components of Code view of kurchen 4+1 model??

Answer:

1. Classes
2. Modules
3. Components

Q58) GOF Design Pattern Templates names??

Answer:

1. Creational Patterns
2. Structural Patterns
3. Behavioral Patterns

Q59) Stock exchange system uses a lot of screens what does the update system for it?

Answer:

A stock exchange system with many screens needs a real-time update mechanism, such as

- Push Notifications
- WebSockets
- Polling

Q60) Write get/set and prefix function with example in programming language?

Answer:

```
public class Example {  
    private int value;  
    public int getValue() {
```

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```
        return value;
    }
    public void setValue(int value) {
        this.value = value;
    }
    public boolean isPositive() {
        return value > 0;
    }
}
```

Q61) Multiple inheritance, association, class/classes, inheritance, Expolation definition??

Answer:

Multiple Inheritance: A class can inherit from more than one class.

Association: A relationship between two classes.

Class/Classes: Blueprints for creating objects.

Inheritance: A class can inherit properties and methods from another class.

Exploration: Analyzing and investigating data or structures.

Q62) Black box testing is different from white box testing?

Answer:

Black box testing focuses on external behavior, while white box testing examines internal structure.

Q63) Write white box testing schemes?

Answer:

✚ Statement Coverage

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- ✚ Branch Coverage
- ✚ Path Coverage
- ✚ Condition Coverage

Q64) Short circuiting in programming language?

Answer:

Short circuiting is when the evaluation of an expression stops as soon as the outcome is determined. For example, in a `&& b`, if `a` is false, `b` is not evaluated because the whole expression will be false regardless of `b`.

Q65) Three quantitative benefits of unit testing

Answer:

1. Finding and fixing bugs early reduces the number of defects in the final product.
2. Detecting issues early speeds up the development process as less time is spent on debugging.
3. Fixing bugs during unit testing is cheaper than fixing them after the product is released.

Q66) What issues can arise if system is design without keeping portability in mind?

Answer:

If a system is designed without portability in mind, it can lead to:

- Compatibility Issues
- Increased Costs
- Limited Market Reach

Q67) Logical errors and it's symptoms

Answer:

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Logical Errors mistakes in the code logic that cause the program to behave incorrectly.

Symptoms

- Incorrect output results.
- Program crashes.
- Unexpected behavior under certain conditions.

Q68) What techniques can be used to design system that is loosely coupled and highly cohesive

Answer:

Use Interfaces and Abstractions: To reduce dependencies between components.

Modular Design: Break the system into small, independent modules.

Encapsulation: Keep data and methods within the same class to increase cohesion.

Q69) Which techniques can be used to design the system without ambiguity from user's point of view

Answer:

- ✚ Ensure that user requirements are precise and unambiguous.
- ✚ Build prototypes to clarify requirements and expectations.
- ✚ Use these to capture detailed user requirements.

Q70) Two areas where the design pattern are used

Answer:

1. Software Development
2. UI Design

Q71) Write the construction activities in development of software.

Answer:

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- ✚ Coding
- ✚ Unit Testing
- ✚ Integration Testing
- ✚ Code Review

Q72) Give example code of "Get/set" functions and "Is" prefix.

Answer:

```
public class Example {  
    private int value;  
    // Get/Set methods  
    public int getValue() {  
        return value;  
    }  
    public void setValue(int value) {  
        this.value = value;  
    }  
    // Is method  
    public boolean isPositive() {  
        return value > 0;  
    }  
}
```

Q73) Write the structure for exception handling code.

Answer:

```
try {  
    // Code that may throw an exception
```

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```
} catch (SpecificExceptionType e) {  
    // Handle the specific exception  
} catch (AnotherExceptionType e) {  
    // Handle another type of exception  
} finally {  
    // Code that will always run, regardless of whether an exception is thrown or not  
}
```

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0316-2965677

2025

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